Farhan Ali

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PROFESSIONAL SUMMARY

Research-oriented Medical Laboratory Technology graduate (3.69 CGPA) with specialized expertise in computational biology, molecular diagnostics, and clinical research. Eager to contribute to innovative research addressing the molecular basis of human diseases and advancing precision medicine in a rigorous academic environment.

EDUCATION

Bachelor of Science in Medical Laboratory Technology (BS MLT)
Khyber Medical University (KMU), Institute of Paramedical Sciences | *Peshawar, Pakistan* | *2021 – 2025*

- **CGPA:** 3.69/4.0 (Ranked among the top 6 students in the class).
- **Academic Excellence:** Achieved a perfect 4.0 GPA across the final two academic semesters (7th and 8th).
- **Thesis Defense:** Successfully defended undergraduate research thesis with formal commendation and high praise from external academic examiners.

RESEARCH & COMPUTATIONAL EXPERIENCE

Undergraduate Clinical Researcher
Khyber Medical University | *Peshawar, Pakistan* | *2025 – 2026*

- Investigated psychosocial challenges, health-related quality of life, and caregiver burden in children with Transfusion-Dependent Beta Thalassemia.
- Designed a cross-sectional study, recruiting 200 clinical child-parent dyads, and processed clinical data using SPSS and R Studio for independent samples t-tests, one-way ANOVA, and Pearson correlation analyses.
- **Publication:** Co-authored a manuscript titled "Psychosocial difficulties, Health-Related Quality of life, And Caregiver Burden in Children with Transfusion-Dependent B-Thalassemia," currently submitted to Thalassemia Reports.

Independent Bioinformatics Developer
Hosted on GitHub | *07/2026 – 08/2026*

- **Clinical NGS Variant Annotator:** Engineered an automated Python-based bioinformatics pipeline to ingest targeted Next-Generation Sequencing (NGS) variant panels, integrating the Ensembl REST API to query international databases and map disease-causing variants to severe hematological disorders.
- **Genomic Sequence Analysis Pipeline:** Developed an automated pipeline utilizing Python and Biopython to integrate BLAST algorithms with NCBI databases for precise gene-finding and structural stability analysis.

TECHNICAL & LABORATORY SKILLS

- **Programming & Biostatistics:** Python (Biopython), R Studio (ggplot2), Linux/Bash Scripting, SPSS v27.
- **Bioinformatics:** Sequence alignment (BLAST, NCBI), genome annotation, 3D protein modeling (AlphaFold, SWISS-MODEL), database mining (UniProt, PDB, ExPASy).
- **Clinical Diagnostics:** PCR-based molecular diagnostics, Hematology (CBC, Hb electrophoresis), Serology (ELISA, Coombs' testing), Microbiology, Blood Banking, and stringent QA/QC protocol management.

HONORS, AWARDS & LEADERSHIP

- **Aspire Leadership Program (Cohort 3):** Selected participant for a globally competitive leadership development program founded by Harvard University Faculty (07/2026 – Present).
- **Prime Minister’s Laptop Scheme:** Awarded a highly competitive, merit-based technology grant recognizing academic excellence in core laboratory sciences (2025).
- **Stoori Da Pakhtunkhwa Program:** Honored by the Elementary and Secondary Education Department for securing the College 1st Position (960/1100 marks) in early education (2021).
- **National Organization for Rare Disorders (NORD):** Active participant engaging in rare disease advocacy and education initiatives (08/2026 – Present).
- **Selected Certifications:** Writing in the Sciences (Stanford University/Coursera), Bioinformatics for Biologists: Introduction to Linux, Bash Scripting, and R (Wellcome Connecting Science).